SN	BCA201 2	Computer Organization	L	T	P	C	Course Type
			3	0	0	3	Theory
Version		BCA201226					
PRE-REQUISITE							
CO-REQUISITE		None					
ANTI-REQUISITE		None					

a. Course objectives-

- To introduce the fundamental concepts of computer organization, digital data representation, Boolean algebra, and combinational circuit design.
- To develop understanding of CPU architecture, instruction execution, instruction formats, and control unit mechanisms.
- To build knowledge of memory hierarchy, memory technologies, cache organization, and storage systems.
- To familiarize students with input/output organization, data transfer mechanisms, interrupts, and DMA concepts.
- To provide foundational exposure to assembly language, pipelining, and the integration of computer system components for performance understanding.

b. Course outcomes-

CO1	Explain computer organization fundamentals, number systems, Boolean algebra, logic gates, and combinational circuits.
CO2	Apply concepts of CPU architecture, instruction cycles, addressing modes, and data representation in computational processes.
CO3	Utilize memory hierarchy concepts, cache mechanisms, and storage technologies to interpret memory organization and performance.
CO4	Analyze input/output organization, interrupt handling, and DMA techniques for efficient data transfer within computer systems.
CO5	Evaluate assembly language concepts, pipelining techniques, and integrated computer architecture for system-level performance improvement.

c. Syllabus

Unit-1	Introduction and Physical Data Handling	Contact Hours: 12
	Computer Organization: What is Computer Organization vs. Computer Architecture, The basic functional units of a computer- Input, Output, Memory, Arithmetic Logic Unit (ALU), and Control Unit, Von Neumann and Harvard architectures. Number Systems and Codes: Binary, octal, decimal, and hexadecimal number systems, Conversions between different number systems, Introduction to binary arithmetic (addition, subtraction): BCD, Gray code, and ASCII codes. Boolean Algebra and Logic Gates: Boolean algebra axioms and theorems, Logic gates: AND, OR, NOT, NAND, NOR, XOR, and XNOR, Truth tables and logic expressions. Combinational Circuits: Introduction to combinational circuits, Designing half-adder and full-adder circuits, Introduction to multiplexers (MUX) and demultiplexers (DEMUX).	
Unit-2	The Central Processing Unit (CPU)	Contact Hours: 12
	Basic CPU Structure: Registers: General purpose and special purpose registers (e.g., Program Counter, Instruction Register, Accumulator), The Instruction Cycle: Fetch, Decode, Execute, and Write-back. Instruction Set Architecture (ISA):What is an instruction set, Instruction formats and types (e.g., zero-address, one-address, two-address), Addressing modes. Data Representation: Signed and unsigned number representation, Floating-point representation (IEEE 754 standard). Hardwired vs. Microprogrammed Control: Understanding the role of the Control Unit, A high-level overview of hardwired control and microprogrammed control.	
Unit-3	Memory Organization	Contact Hours: 10
	Introduction to Memory: Memory hierarchy: Registers, Cache, Main Memory (RAM), and Secondary Storage, Characteristics of memory systems: speed, size, cost. Primary Memory (Main Memory): RAM (Static and Dynamic), ROM (PROM, EPROM, EEPROM). Cache Memory: The concept of cache memory, Locality of reference, Cache mapping techniques (direct, associative, set-associative). Secondary Storage: Introduction to Hard Disk Drives (HDD) and Solid-State Drives (SSD), Memory-mapped I/O and isolated I/O.	
Unit-	Input/Output Organization	Contact Hours: 5

4		
	I/O Devices: Types of I/O devices (e.g., keyboard, mouse, monitor, printer), The role of I/O interfaces and controllers, Programmed I/O: Understanding how the CPU controls data transfer. Interrupt-driven I/O: The concept of interrupts, Handling multiple interrupts. Direct Memory Access (DMA): The need for DMA, How DMA controllers bypass the CPU for high-speed data transfer.	
Unit-5	**Assembly Language and Advanced Concepts**	**Contact Hours: 6**
	Assembly Language: The role of an assembler, A simple example of assembly language instructions (e.g., MOV, ADD), The relationship between assembly language and machine code. Pipelining: The concept of instruction pipelining for performance improvement, Basic pipeline stages. Course Review: A comprehensive review of all modules, Connecting the concepts of digital logic, CPU, memory, and I/O to form a complete picture of a computer system.	

d. Textbooks/ reference books

TEXTBOOKS

T1: Computer System Architecture by M. Morris Mano, Pearson Education.1992
T2: Computer Organization and Architecture: Designing for Performance by William Stallings, Pearson Education.2023
T3: Computer Organization by Carl Hamacher, Zvonko Vranesic, and Safwat Zaky, McGraw Hill Education.2011

REFERENCE BOOKS

R1: Fundamentals of Digital Logic with VHDL Design by Stephen Brown and Zvonko Vranesic, McGraw Hill Education. 2019
R2: Digital Logic and Computer Design by M. Morris Mano, Pearson Education. 1979
R3: Computer Architecture: A Quantitative Approach by John L. Hennessy and David A. Patterson, Morgan Kaufmann.2017

e. Assessment Pattern - Internal and External

The performance of students is evaluated as follows:

Theory Courses: Evaluation Component:

	Theory- Mandatory Graded	
Component	**Continuous Internal Assessment (CIA)**	**End Semester Examination (ESE)**
Marks	**50**	**50**
Total Marks	**100**	

S. No	**Components**	**Type of Assessment**	**Marks**	**Final Weightage Assessment**
1	**Continuous Internal Assessment (CIA)**	First Assessment test (FAT)	30	15%
2		Second Assessment test (SAT)	30	15%
3		Assignment/ Quiz/ Surprise test/ Mini Project/ Presentation/ Group Discussion etc.	20	20%
4	**End Semester Examination (ESE)**	End Semester Examination (ESE))	50	50%

f. CO-PO Mapping:

CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PSO1**	**PSO2**
CO1	3	2	2	1	–	–	–	1	–	1	2	1
CO2	3	3	2	2	–	–	–	1	1	2	3	2
CO3	3	3	2	2	–	–	–	1	1	2	3	2
CO4	3	3	3	2	–	1	–	1	1	2	3	2
CO5	3	3	3	3	1	1	1	2	2	3	3	3